stability they need. You also need to make sure that you overlap the access point coverage areas sufficiently to offer the maximum possible network speeds to your users.

You will obviously need a lot of Wi-Fi-enabled equipment to make use of your WLAN setup, and if this equipment isn't already deployed, this cost needs to be factored in as well.

The Site Survey

A proper Radio Frequency (RF) site survey is a requisite, and it can be done in two ways—as a task left to the firm you hire to deploy WLAN, or done by your in-house IT deployment team. Of course, smaller businesses could have a systems engineer using one access point and a laptop to set up and walk around the site checking signal strengths. Whatever the case, your WLAN deployment needs planning, as anything, from a wall to machinery can disrupt the wireless signals.

You must also consider actual working conditions when planning your WLAN setup. For instance, a warehouse where a site survey is done when there is less activity and no goods are stored; after deployment, however, you may find that running machinery, or even piled up goods, can interfere with the wireless signals.

You should recognise the dynamics of your environment. As explained above, a warehouse may require different deployment tactics than a regular office. In an office, users will be at their desks most of the time. However, a regular office will also have cubicles, walls and other obstructions that will adversely affect the signal.

A Seamless WLAN

A seamless WLAN setup is one where users can stroll across the covered area, and at the same time have constant access to the network. This is easier said than done, and is a lot more complicated than just setting up access points to cover the desired area. The 802.11 specification offers 14 different channels, or frequencies, at which access points can be set. The problem is, these frequencies are too close together—a difference of just five MHz between channels—and interference is caused. To overcome this inter-

Access Point Antennas

Perhaps the most ignored feature when choosing an access point, antennas should be given considerable thought. Most access points come with omni-directional antennas, which means that the access point sends out a signal in the form of a sphere all around itself. This is especially useful when placing an access point in the centre of a smaller office, as it can cover the entire office area easily. However, it is when we reach larger offices—where numerous access points are needed—that directional access point antennas may offer you some relief.

Directional antennas also offer some relief for those obsessed with security, as you can ensure that all your access points offer a known and fixed coverage area. Directional antenna-based access points could be set to ensure that no one ever gets a signal outside the office walls, and thus cannot hack into the network. Directional antennas also help you plan and position your access points better, so as to offer maximum coverage. Their biggest advantage, however, is the fact that they offer greater bandwidth and range, as a result of being focussed in a single direction.

When deploying a WLAN in a large multi-storeyed office, uni-directional antennas will be needed to ensure that there is no overlap of signals of access points running on the same channel, thus breaking the seamless network.